INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Worksheet 1 Solutions

1. A pharmacy needs a lead pharmacist and a backup pharmacist, chosen from its 5 pharmacists, and 3 pharmacy technicians with no distinct roles, chosen from its 8 technicians. In how many ways can these positions be filled?

Solution: The lead and backup pharmacists have different roles, so order matters, and there are

$$ {}_5P_2 = \frac{5!}{(5-2)!} = \frac{5 \cdot 4 \cdot 3!}{3!} = 5 \cdot 4 = 20 $$

ways to choose them. The technicians have no distinct roles, so order does not matter, and there are

$$ {}_8C_3 = \frac{8!}{3!\,(8-3)!} = \frac{8 \cdot 7 \cdot 6 \cdot 5!}{3! \cdot 5!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = \frac{336}{6} = 56 $$

ways to choose them. By the Fundamental Principle of Counting, the positions can be filled in

$$ {}_5P_2 \cdot {}_8C_3 = 20 \cdot 56 = 1{,}120 $$

ways.

2. The hospital cafeteria's lunch special is either a bowl of soup with a roll, or a salad. There are 4 soups and 3 kinds of rolls to choose from. A salad comes with 2 different toppings, chosen from 6 toppings. How many different lunch specials are possible?

Solution: We split the lunch specials into two cases.

Case 1: Soup with a roll. There are 4 soups and 3 rolls, so by the Fundamental Principle of Counting there are $4 \cdot 3 = 12$ such specials.

Case 2: A salad. The 2 toppings have no order, so there are

$$ {}_6C_2 = \frac{6!}{2!\,(6-2)!} = \frac{6 \cdot 5 \cdot 4!}{2! \cdot 4!} = \frac{6 \cdot 5}{2 \cdot 1} = 15 $$

such specials.

No special is both a soup and a salad, so the cases are disjoint, and by the Addition Principle there are $12 + 15 = 27$ possible lunch specials.

3. A committee of 3 will be chosen from 5 nurses and 4 doctors, and the committee members have no distinct roles. The committee must have more nurses than doctors. How many different committees are possible?

Solution: A committee of 3 with more nurses than doctors has either 3 nurses or 2 nurses and 1 doctor.

Case 1: 3 nurses. There are

$$ {}_5C_3 = \frac{5!}{3!\,(5-3)!} = \frac{5 \cdot 4 \cdot 3!}{3! \cdot 2!} = \frac{5 \cdot 4}{2 \cdot 1} = 10 $$

such committees.

Case 2: 2 nurses and 1 doctor. There are

$$ {}_5C_2 = \frac{5!}{2!\,(5-2)!} = \frac{5 \cdot 4 \cdot 3!}{2! \cdot 3!} = \frac{5 \cdot 4}{2 \cdot 1} = 10 $$

ways to choose the nurses and

$$ {}_4C_1 = \frac{4!}{1!\,(4-1)!} = \frac{4 \cdot 3!}{1 \cdot 3!} = 4 $$

ways to choose the doctor, so by the Fundamental Principle of Counting there are $10 \cdot 4 = 40$ such committees.

The cases are disjoint, so by the Addition Principle there are $10 + 40 = 50$ possible committees.

4. A coin is tossed 8 times. How many of the possible sequences of heads and tails contain at least 2 heads?

Solution: We use complements. By the Fundamental Principle of Counting, there are $2^8 = 256$ sequences in all. The complement of "at least 2 heads" is "0 or 1 head." There is

$$ {}_8C_0 = \frac{8!}{0!\,(8-0)!} = \frac{8!}{1 \cdot 8!} = 1 $$

sequence with no heads, and there are

$$ {}_8C_1 = \frac{8!}{1!\,(8-1)!} = \frac{8 \cdot 7!}{1 \cdot 7!} = 8 $$

sequences with exactly one head. Therefore, the number of sequences with at least 2 heads is

$$ 256 - 1 - 8 = 247. $$

5. A hospital's Wi-Fi password has 4 characters, each of which is a lowercase letter (a–z) or a digit (0–9). Letters and digits may be repeated. How many possible passwords contain at least one digit?

Solution: We use complements. Each character can be any of 26 letters or 10 digits, so by the Fundamental Principle of Counting there are $36^4 = 1{,}679{,}616$ passwords in all. A password with no digits is made up of 4 letters, and there are $26^4 = 456{,}976$ of these. Therefore, the number of passwords with at least one digit is

$$1{,}679{,}616 - 456{,}976 = 1{,}222{,}640.$$

6. A team of 5 will be chosen from 8 nurses and 6 medical assistants, and the team members have no distinct roles. How many different teams include at least one nurse and at least one medical assistant?

Solution: We use complements. There are

$${}_{14}C_5 = \frac{14!}{5!\,(14-5)!} = \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9!}{5! \cdot 9!} = \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{240{,}240}{120} = 2{,}002$$

teams in all. A team fails to include at least one nurse and at least one medical assistant exactly when it is all nurses or all medical assistants. There are

$${}_8C_5 = \frac{8!}{5!\,(8-5)!} = \frac{8 \cdot 7 \cdot 6 \cdot 5!}{5! \cdot 3!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = \frac{336}{6} = 56$$

teams of all nurses and

$${}_6C_5 = \frac{6!}{5!\,(6-5)!} = \frac{6 \cdot 5!}{5! \cdot 1!} = 6$$

teams of all medical assistants, and these cases are disjoint. Therefore, the number of teams with at least one of each is

$$2{,}002 - 56 - 6 = 1{,}940.$$

7. A hospital surveyed 120 of its employees. Of these, 70 had received a flu shot this season, 45 had received a COVID-19 booster, and 30 had received both.

(a) How many of the employees had received at least one of the two?

Solution: Let F be the set of employees who had received a flu shot and C the set of employees who had received a COVID-19 booster. By the Inclusion-Exclusion Principle,

$$|F \cup C| = |F| + |C| - |F \cap C| = 70 + 45 - 30 = 85.$$

(b) How many had received neither?

Solution: These employees make up the complement $(F \cup C)^c$, so there are $120 - 85 = 35$ of them.

(c) How many had received exactly one of the two?

Solution: Of the 85 employees who had received at least one of the two, 30 had received both, so $85 - 30 = 55$ had received exactly one.

8. The lockers in a clinic's staff room open with 3-digit codes, such as 070. Each digit can be any of 0 through 9, and digits may be repeated. How many codes begin with 7 or end with 7?

Solution: Let A be the set of codes that begin with 7 and B the set of codes that end with 7. In A, the other two digits can be anything, so $|A| = 10 \cdot 10 = 100$. In the same way, $|B| = 100$. In $A \cap B$, only the middle digit can vary, so $|A \cap B| = 10$. By the Inclusion-Exclusion Principle,

$$|A \cup B| = |A| + |B| - |A \cap B| = 100 + 100 - 10 = 190.$$

We can check this using complements: there are $1{,}000$ codes in all, and $9 \cdot 10 \cdot 9 = 810$ of them neither begin nor end with 7, so $1{,}000 - 810 = 190$.

9. A hospital unit needs a charge nurse and 4 staff nurses with no distinct roles for the night shift, all chosen from its 12 nurses. No nurse can fill more than one position. Two of the nurses, Dana and Eli, carpool to work, so either both of them work the night shift or neither of them does. In how many ways can the night shift be staffed?

Solution: We split the shifts into cases. The cases are disjoint, so we add them at the end using the Addition Principle.

Case 1: Neither Dana nor Eli works. Choose the charge nurse from the other 10 nurses, in 10 ways, and then the 4 staff nurses from the remaining 9, in

$$ {}_9C_4 = \frac{9!}{4!\,(9-4)!} = \frac{9\cdot 8\cdot 7\cdot 6\cdot 5!}{4!\cdot 5!} = \frac{9\cdot 8\cdot 7\cdot 6}{4\cdot 3\cdot 2\cdot 1} = \frac{3{,}024}{24} = 126 $$

ways, for $10\cdot 126 = 1{,}260$ shifts.

Case 2: Both work, and one of them is the charge nurse. There are 2 ways to choose which of them is the charge nurse. The other one is a staff nurse, and the remaining 3 staff nurses are chosen from the other 10 nurses in

$$ {}_{10}C_3 = \frac{10!}{3!\,(10-3)!} = \frac{10\cdot 9\cdot 8\cdot 7!}{3!\cdot 7!} = \frac{10\cdot 9\cdot 8}{3\cdot 2\cdot 1} = \frac{720}{6} = 120 $$

ways, for $2\cdot 120 = 240$ shifts.

Case 3: Both work, and neither is the charge nurse. Choose the charge nurse from the other 10 nurses, in 10 ways. Dana and Eli are 2 of the staff nurses, and the other 2 are chosen from the remaining 9 nurses in

$$ {}_9C_2 = \frac{9!}{2!\,(9-2)!} = \frac{9\cdot 8\cdot 7!}{2!\cdot 7!} = \frac{9\cdot 8}{2\cdot 1} = 36 $$

ways, for $10\cdot 36 = 360$ shifts.

Therefore, there are $1{,}260 + 240 + 360 = 1{,}860$ ways to staff the night shift.

10. A committee of 5 will be chosen from 7 nurses and 6 doctors, and the committee members have no distinct roles. The committee must include at least one nurse and at least one doctor. In addition, Nurse Kim and Dr. Patel refuse to serve on the committee together. How many different committees are possible?

Solution: First ignore Kim and Patel. There are

$$_{13}C_5 = \frac{13!}{5!\,(13-5)!} = \frac{13\cdot 12\cdot 11\cdot 10\cdot 9\cdot 8!}{5!\cdot 8!} = \frac{13\cdot 12\cdot 11\cdot 10\cdot 9}{5\cdot 4\cdot 3\cdot 2\cdot 1} = \frac{154{,}440}{120} = 1{,}287$$

committees in all. A committee without at least one nurse and at least one doctor is all nurses or all doctors. There are

$$_7C_5 = \frac{7!}{5!\,(7-5)!} = \frac{7\cdot 6\cdot 5!}{5!\cdot 2!} = \frac{7\cdot 6}{2\cdot 1} = 21$$

committees of all nurses and

$$_6C_5 = \frac{6!}{5!\,(6-5)!} = \frac{6\cdot 5!}{5!\cdot 1!} = 6$$

of all doctors, so using complements, there are

$$1{,}287 - 21 - 6 = 1{,}260$$

committees with at least one nurse and at least one doctor.

Now remove the committees with both Kim and Patel. Such a committee is decided by choosing the other 3 members from the remaining 11 people, in

$$_{11}C_3 = \frac{11!}{3!\,(11-3)!} = \frac{11\cdot 10\cdot 9\cdot 8!}{3!\cdot 8!} = \frac{11\cdot 10\cdot 9}{3\cdot 2\cdot 1} = \frac{990}{6} = 165$$

ways. Every one of these committees already has a nurse (Kim) and a doctor (Patel), so all 165 were counted in the $1{,}260$. Therefore, the number of possible committees is

$$1{,}260 - 165 = 1{,}095.$$